

Climbing Injury Prevention & Management

A Course for Physiotherapists

MODULE 4: ASSESSING & DIAGNOSING COMMON CLIMBING INJURIES

4.1 Purpose of This Module

Modules 2 and 3 built the anatomical, biomechanical, and epidemiological foundation for climbing injury. This module puts that foundation to clinical use: a site-by-site framework for assessing and differentiating the injuries most likely to present in a climbing patient, structured around six regions — fingers, wrist, elbow, shoulder, knee, and ankle/foot. Each section pairs the climbing-specific mechanism with the physical examination findings and imaging pathway that help distinguish between the realistic differentials for that site. This is the direct clinical bridge into Module 5's treatment and rehabilitation content.

4.A Fingers & Hand Injuries

Finger injuries in climbers

Building on Module 3, three finger presentations account for most climbing-related finger pathology. Distinguishing between them clinically shapes both the imaging pathway and the management plan.

PULLEY INJURY	
Onset	Acute — often a single “pop” during a crimped or dynamic move.
Pain location	Focal, volar, at A2/A4 level.
Provocation	Pain and possible bowstringing on resisted flexion.
Key exam sign	Bowstringing test: resisted flexion with PIP ~20° / DIP ~40° counter-flexion. Palpation in crimp position (85° PIP flexion; 5° DIP extension).
First-line imaging	Dynamic ultrasound (gold standard); MRI if inconclusive.

FLEXOR TENOSYNOVITIS	
Onset	Gradual — builds with increased training volume/intensity.
Pain location	Diffuse volar swelling along the tendon sheath. Tender on palpation around A2 pulley.
Provocation	Pain with gripping/loading; may ease then return with rest.
Key exam sign	Diffuse tenderness with generalised mild oedema along the sheath without focal bowstringing.
First-line imaging	Ultrasound can show sheath thickening/effusion (“halo phenomenon” in transverse view); often clinical diagnosis.

CAPSULITIS / SYNOVITIS

Onset	Gradual — often linked to asymmetric loading or faulty gripping mechanics.
Pain location	Localised to the PIP or DIP joint line / capsule distribution.
Provocation	Often eases with warm-up and mid-range movement.
Key exam sign	Joint-line tenderness; may have subtle or pronounced effusion.
First-line imaging	Usually clinical; imaging reserved for atypical or persistent cases.

COLLATERAL LIGAMENT SPRAIN

Onset	Side-loading of a finger, twisting while crimping or catching a hold, esp. pockets.
Pain location	Pain on one side of a finger joint, swelling, instability.
Provocation	Tenderness over collateral ligament.
Key exam sign	Pain with valgus or varus stress; compare laxity with the opposite hand; test in 0° extension and 30° flexion.
First-line imaging	Plain radiographs (AP, lateral, oblique) to exclude avulsion fractures or associated joint injury. Ultrasound can demonstrate ligament disruption dynamically. MRI reserved for equivocal cases, high-level athletes, or suspected complete tears requiring surgical planning.

VOLAR PLATE INJURY

Onset	Sudden hyperextension of the PIP joint.
Pain location	Pain on the palmar aspect of the PIP joint with swelling.
Provocation	Pain during passive extension and gripping.
Key exam sign	Swelling and bruising around the PIP joint; possible extension lag or flexion deformity in more severe injuries.
First-line imaging	Plain radiographs (AP, lateral, oblique) to exclude an avulsion fracture or PIP subluxation/dislocation. Ultrasound or MRI if diagnosis is uncertain, persistent pain, or associated soft-tissue injury is suspected.

Pulley Injury: Grading and Assessment in Detail

The bowstringing test remains the most widely cited clinical sign of pulley rupture, performed with the finger's PIP joint at approximately 20° and DIP at approximately 40° of flexion while the examiner applies resistance against the fingertip (the “counter-flexion” technique) — increasing resistance increases any bowstring effect, making it easier to detect. It is worth noting, however, that systematic review evidence has not established formal sensitivity or specificity values for this test in isolation, so clinical findings should be interpreted alongside imaging rather than used to rule pulley injury in or out on their own.

Grade	Injury	Typical Management
I	Pulley strain — no tendon-bone dehiscence.	Conservative: taping, activity modification.
II	Complete A4 rupture, or partial A2/A3 rupture.	Conservative: taping, graded reloading.
III	Complete rupture of A2 or A3.	Conservative: thermoplastic ring + taping.
IV	Multiple pulley rupture, or single rupture with lumbrical or collateral ligament involvement.	Surgical reconstruction generally indicated.

CLINICAL NOTE

Dynamic ultrasound is considered the reference standard for diagnosing pulley rupture and grading severity, since it allows real-time assessment of tendon-to-bone distance during resisted flexion. A healthy pulley results in a tendon to bone distance of 2mm, a torn one 4–6mm; MRI is a useful second-line study when ultrasound is inconclusive or when associated structures (lumbrical, collateral ligaments) need clearer visualisation.

Hand Injuries in Climbers

Although flexor pulley injuries receive the greatest attention in climbing medicine, several other structures within the hand are frequently injured. These conditions often present with symptoms that mimic pulley injuries, making accurate assessment essential. Careful history taking, knowledge of the mechanism of injury, and a thorough clinical examination are often sufficient to distinguish these injuries and guide appropriate management.

The intrinsic muscles of the hand, collateral ligaments, joint capsules, digital nerves and flexor tendon sheath can all become overloaded during climbing. These injuries are particularly common during dynamic movements, two- and three-finger pocket climbing, aggressive crimping, gastons and twisting movements where individual fingers are subjected to different loads. Although many of these injuries respond well to conservative management, delayed diagnosis can prolong recovery and contribute to persistent pain or reduced grip performance.

Common Hand Injury in Climbers: Lumbrical Strain (Lumbrical Shift Syndrome)

Lumbrical injuries are among the more distinctive soft tissue injuries encountered in climbers. Unlike most muscles, the lumbricals originate from the flexor digitorum profundus tendons rather than directly from bone, before inserting into the extensor mechanism on the back of the fingers. This unique anatomy allows them to coordinate finger movement and contribute to fine motor control, but also makes them vulnerable when adjacent fingers are loaded in different positions.

The classic injury mechanism occurs during climbing on two- or three-finger pockets. While one finger remains deeply flexed inside the pocket, another may become relatively extended or unloaded. This creates tension between adjacent flexor tendons, placing a shearing force across the lumbrical muscle. This phenomenon, commonly referred to as the Quadriga effect, is unique to the anatomy of the hand and explains why lumbrical injuries are seen far more frequently in climbers than in most other athletes.

A second, equally common mechanism — sometimes referred to informally as the “pinkie drop” — occurs during the three-finger drag grip, where the little finger is deliberately kept off the hold while the middle three fingers bear load.

Because the little finger's flexor tendon stays relatively slack while its neighbours are fully loaded, the same relative-extension shearing mechanism responsible for pocket-related strain occurs here too, independent of pocket use entirely. This has a direct rehabilitation implication: where the three-finger drag is the primary or contributing mechanism, return-to-climbing progressions need to specifically reintroduce and load that exact grip pattern in a controlled, graded way, rather than assuming general open-hand reloading is sufficient.

Patients typically report pain deep within the palm, most commonly at the base of the ring finger, rather than directly over the finger joints or pulleys. Symptoms may initially present as a dull ache but become sharp when the injured finger is loaded independently or when pocket climbing is attempted again.

Clinical Assessment

Several examination findings help distinguish a lumbrical injury from pulley or flexor tendon pathology.

- Pain is located deep within the palm rather than over the finger.
- Symptoms increase when one finger is passively extended while adjacent fingers remain flexed.
- Pocket climbing is significantly more painful than open-hand gripping.
- Direct palpation between the flexor tendons reproduces symptoms.
- Commonly associated with FDP strains and co-existing forearm pain on palpation and contraction.

Because pulley injuries, flexor tendon strains and collateral ligament injuries may produce similar symptoms, clinicians should always consider these differential diagnoses during assessment.

Management Principles

Most lumbrical injuries respond well to conservative management. Early treatment focuses on reducing excessive loading while maintaining finger mobility. Climbers should temporarily avoid pocket climbing and any movement that reproduces sharp pain, but complete immobilisation is rarely required.

As symptoms improve, rehabilitation progresses towards restoring intrinsic hand strength, independent finger control and gradual reloading through open-hand positions before returning to pockets. Progressive exposure to climbing-specific loading remains the most effective strategy for returning athletes safely to performance.

A simple technique-correction point is worth passing on to coaches and climbers alike as prevention: actively keeping the proximal phalanx of the little finger adjacent to the ring finger — rather than allowing it to drift — reduces the relative shearing force between the two flexor tendons that underlies this injury. This is an easily coached strategy worth introducing alongside any rehabilitation programme, particularly for climbers who habitually favour pocket use or the three-finger drag.

CLINICAL NOTE

Climbers often describe any finger pain as a “pulley injury.” Pain located in the palm rather than over the finger, especially after climbing on two- or three-finger pockets, should immediately raise suspicion of a lumbrical strain. Accurate diagnosis prevents unnecessary pulley rehabilitation and allows a more targeted return-to-climbing programme.

4.B Wrist

Wrist pain in climbers is most often ulnar-sided, reflecting the crimped, ulnar-deviated hand positions described in Module 2. The three presentations below account for the large majority of climbing-related wrist complaints, alongside the hamate and TFCC anatomy already covered.

TFCC INJURY	
Typical mechanism	Ulnar-deviated loading — sidepulls, underclings, big open-hand slopers.
Pain location	Ulnar-sided wrist pain, may click.
Key provocation tests	TFCC compression test, supination lift test, ulnar impingement test. (See the hosted lesson page for linked video demonstrations.)
Imaging note	Provocative tests have limited sensitivity/specificity alone; correlate with MRI or arthroscopy if suspicion is high.

HAMATE HOOK STRESS/FRACTURE	
Typical mechanism	Repetitive ulnar-side crimping load transmitted through flexor tendons to the hook / repetitive intense underclings.
Pain location	Diffuse, deep palmar/ulnar wrist pain over the hook of hamate.
Key provocation tests	Pain with resisted flexion of ring/little fingers in ulnar-deviated wrist.
Imaging note	Plain radiograph first; CT or MRI if radiograph unclear — diffuse presentation is easy to miss.

DE QUERVAIN'S TENOSYNOVITIS	
Typical mechanism	Vertical pinches and closed crimping loading the thumb into abduction/extension.
Pain location	Radial-sided wrist pain over the first dorsal compartment.
Key provocation tests	Finkelstein's test. (See the hosted lesson page for a linked video demonstration.)
Imaging note	Usually a clinical diagnosis; imaging rarely required.

FCU TENDINOPATHY

Typical mechanism	Repetitive forceful wrist flexion and ulnar deviation during overhang climbing, lock-offs, underclings, gastons, and campus board training.
Pain location	Localised pain over the FCU tendon at the pisiform and/or medial aspect of the distal forearm.
Key provocation tests	Pain reproduced with resisted wrist flexion and ulnar deviation; pain on passive wrist extension; tenderness on palpation of the FCU tendon near the pisiform or along its course.
Imaging note	Ultrasound is useful for demonstrating tendon thickening, hypoechogenicity, and Doppler evidence of neovascularisation. MRI may be indicated if tendon tear or alternative pathology is suspected.

ULNAR NERVE ENTRAPMENT

Typical mechanism	Prolonged or repetitive elbow flexion (cubital tunnel) or sustained pressure through the hypothenar eminence during climbing (Guyon's canal). May also result from repetitive gripping, dynamic moves, or direct trauma to the medial elbow or palm, or from increased tone of the FCU.
Pain location	Medial elbow (cubital tunnel) or hypothenar region of the hand (Guyon's canal), often accompanied by numbness or tingling in the little finger and ulnar half of the ring finger. Symptoms may radiate into the forearm.
Key provocation tests	Positive Tinel's sign over the cubital tunnel or Guyon's canal; sustained pressure over Guyon's canal; sensory testing of the ulnar digits; weakness of finger abduction/adduction. Positive ULTT2.
Imaging note	Diagnosis is primarily clinical. Nerve conduction studies and electromyography help confirm and localise the lesion. Ultrasound can demonstrate nerve enlargement or compression, while MRI is reserved for suspected space-occupying lesions or atypical presentations.

CLINICAL NOTE

Provocative wrist tests — including those for TFCC pathology — have consistently shown limited diagnostic accuracy in isolation in the physiotherapy literature. Use a cluster of findings (mechanism, palpation, and provocation tests together) to build clinical suspicion, and correlate with imaging before committing to a specific diagnosis, particularly for TFCC and hamate pathology where presentations can be diffuse and easy to miss.

Dorsal wrist impingement — pain on wrist extension, often provoked by mantling moves or crimping with the wrist extended — is a further differential worth considering alongside TFCC pathology, since both can present with dorsal or dorsoulnar-sided pain on loaded extension.

4.C Elbow

Elbow pain in climbers is most frequently medial, reflecting the combined demands of sustained gripping and the pronated pulling position required on the wall. Lateral-sided presentations are less common but remain an important differential, particularly when a presumed epicondylalgia fails to respond to appropriate treatment.

Feature	Medial Epicondylalgia	Lateral Epicondylalgia	Radial Tunnel / PIN Entrapment
Structure involved	Common flexor-pronator origin (flexor carpi radialis, pronator teres).	Common extensor origin (extensor carpi radialis brevis).	Posterior interosseous nerve as it passes through the supinator.
Mechanism in climbers	Sustained gripping combined with the pronated pulling position on the wall.	Less common; associated with heavy lock-off and pressing movements.	Can mimic lateral epicondylalgia; often overlooked.
Key exam finding	Tenderness at the medial epicondyle; pain with resisted wrist flexion/pronation.	Tenderness at the lateral epicondyle; pain with resisted wrist extension (Cozen's test).	Tenderness more distal, over the supinator; pain with resisted supination; deeper, achier quality.
Distinguishing clue	Most frequent elbow presentation in climbers; overuse pattern, gradual onset.	Less frequent in climbers than medial-sided pain.	Absence of focal epicondylar tenderness despite "lateral elbow pain" history; consider if epicondylalgia treatment fails.

4.D Shoulder

Shoulder pain is disproportionately common in more experienced and higher-grade climbers, and presents across a recognisable set of overuse and, less frequently, acute presentations. A standard shoulder examination battery adapted for climbers typically includes:

- Active and passive range of motion, including glenohumeral internal/external rotation.
- Scapular kinematics and stability: clinical observation of scapulohumeral rhythm during active elevation, assessing for deficient posterior tilt or upward rotation, alongside the upper trapezius dominance and serratus anterior/lower trapezius underactivity described in Module 3.
- Instability testing (apprehension test) given the dynamic, sometimes forceful overhead loading involved in climbing moves.
- Rotator cuff assessment: Jobe's test (empty can), external rotation strength, lift-off test, and lag signs.
- Subacromial impingement: Neer's test and the modified Hawkins-Kennedy test.
- Long biceps tendon and superior labral assessment: O'Brien's test, alongside a general awareness that no single clinical test reliably confirms a SLAP lesion in isolation — a cluster of positive findings raises suspicion rather than any one test confirming the diagnosis.
- Acromioclavicular joint: local tenderness and the cross-body adduction test.

Whenever feasible, clinical assessment should be augmented by video analysis of the athlete in motion on the wall, as this often reveals compensatory patterns that clinic-based manual muscle testing may miss. Scapular dyskinesis and excessive recruitment of the rhomboids or upper trapezius are frequently more apparent during loaded, dynamic tasks, providing a clearer diagnostic bridge between table-based findings and actual climbing performance.

CLINICAL NOTE

MRI studies of experienced climbers have found a very high background rate of labral, biceps tendon, and cartilage abnormalities even in shoulders with no current symptoms — in one cohort, every climber studied had multiple MRI abnormalities, yet only a minority of shoulders were symptomatic at the time of scanning. This means imaging findings in a climber's shoulder should be interpreted cautiously and correlated closely with the clinical picture, rather than assumed to explain current symptoms on their own.

4.E Knee

Knee injuries in climbers cluster around four recognisable mechanisms — high step, drop knee, heel hook, and ground fall — each with a reasonably distinct injury profile, reflecting the rotational and shear forces described in Module 3.

Technique / Mechanism	Typical Structure(s) at Risk	Key Assessment Points
High step	Medial meniscus — knee fully flexed and weight-bearing while the tibia/femur rotate.	Joint-line tenderness; McMurray's test; assess for effusion.
Drop knee	Medial meniscus and MCL, via a similar high-flexion rotational load.	As above; also check for pain on passive internal/external rotation at 90° flexion as well as on a valgus stress test and a McMurray with an abduction/external rotation component.
Heel hook	Iliotibial band (sprain occurs almost exclusively via this mechanism), biceps femoris (hamstring), lateral collateral ligament, popliteus, lateral meniscus, posterior cruciate ligament.	Lateral knee palpation, Ober's test for ITB; resisted knee flexion and posterolateral thigh palpation for hamstring; posterior drawer for PCL if hyperflexion/hyperextension mechanism reported.
Ground fall	ACL, multiligament knee injury, complex/bilateral meniscal tears — higher-energy, less climbing-technique-specific pattern.	Lachman's test, posterior drawer, varus/valgus stress testing; have a lower threshold for onward imaging given the higher-energy mechanism.

Ground-fall knee injuries warrant particular caution: evidence from bouldering-specific case series suggests these tend to be higher-energy and more structurally complex than technique-related injuries, with a substantial proportion involving multiple structures (meniscus, ligaments) simultaneously rather than an isolated single-structure injury.

Worth a brief mention alongside the heel hook mechanism: “rocking” onto a high heel hook drives the hip into a combination of deep flexion and external rotation — a position pattern already associated with femoroacetabular impingement and labral irritation in athletes more broadly. This hasn't been specifically studied in climbers, but the mechanical logic is directly analogous to the high step position already discussed above, and hip pain accompanying or following a heel-hook-heavy session is worth considering alongside the knee-focused differentials in the table.

4.F Ankle & Foot

The foot and ankle sit at an unusual intersection in climbing medicine: acute, high-energy trauma from ground falls at one end, and chronic, footwear-driven overuse deformity at the other. Both categories are common enough that it is

worth screening for each specifically, since the two rarely present in quite the same way.

Presentation	Mechanism	Assessment Notes
Lateral ankle sprain	Tight, downturned shoes hold the foot in a relatively supinated position; a loss of purchase during smearing or edging can drive the ankle into inversion.	Anterior drawer and talar tilt tests; recurrent instability is common in climbers and worth asking about specifically.
Forefoot/toe overuse (hallux valgus, hammer/claw toes, subungual haematoma)	Chronic compression from tight, downturned, asymmetric climbing shoes concentrating pressure on the great toe.	Visual inspection of foot shape and nails; ask about typical shoe size relative to street shoe size.
Retrocalcaneal bursitis / Achilles tendinopathy / Haglund's deformity	Heel counter pressure from tight-fitting shoes, aggravated by heel-hooking.	Palpate posterior heel; pain with toe-off or slab-climbing positions.
Calcaneus / talus fracture, osteochondral talar lesion	Ground falls — acute, high-energy axial loading through the heel.	Any fall-related heel/ankle pain with inability to weight-bear warrants imaging; osteochondral lesions may present more insidiously after an initial fall.

CLINICAL NOTE

Foot and ankle injuries are reported to account for close to half of all acute climbing injuries in some cohorts, yet receive far less attention in the climbing-medicine literature and in athlete self-reporting than upper-limb injuries. Actively screening climbing patients for foot and ankle symptoms — rather than waiting for them to be volunteered — is a reasonable practice given this reporting gap.

Summary

A structured, site-by-site assessment approach — grounded in the climbing-specific mechanisms established in Modules 2 and 3 — allows the realistic differentials at each region to be narrowed efficiently: pulley injury, tenosynovitis, and capsulitis in the fingers; TFCC, hamate, and De Quervain's pathology at the wrist; medial and lateral elbow tendinopathy; the shoulder's overuse-dominant pattern; the knee's technique-specific mechanisms; and the ankle/foot's dual acute-versus-overuse picture. With diagnosis established, Module 5 turns to evidence-based treatment and rehabilitation progressions for each of these presentations.

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